

Towards fast and side-channel resistant implementations of Hamming Quasi-Cyclic

A case study on ARM Cortex-M4 devices

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May 4, 2023

Overview

1. Introduction
2. Background
   - Polynomial Multplication
   - Coding theory
   - Hamming Quasi-Cyclic
3. Implementation
   - Polynomial multiplication
   - Masking
4. Experimental results
   - Performance
   - Side-channel resistance
   - Comparison with other cryptosystems
5. Conclusions

Introduction

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The target of this thesis is Hamming Quasi-Cyclic

Goals

- High performance implementation for Cortex-M4
- Protection against timing attacks [GHJ+22, WBB+22]
- Protection against power attacks [SRSW20, SHR+22]

Roadmap

- Evaluate various polynomial multiplication algorithms
- Assess the constant-time behavior of all components
- Implement a masking scheme for the PKE primitives
- Compare the performance with the other Round 4 candidates

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Background

Polynomial multiplication 101

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$$a(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$
$$b(x) = b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n$$

Our goal: calculate:

$$c(x) = a(x) \cdot b(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_{2n} x^{2n}$$

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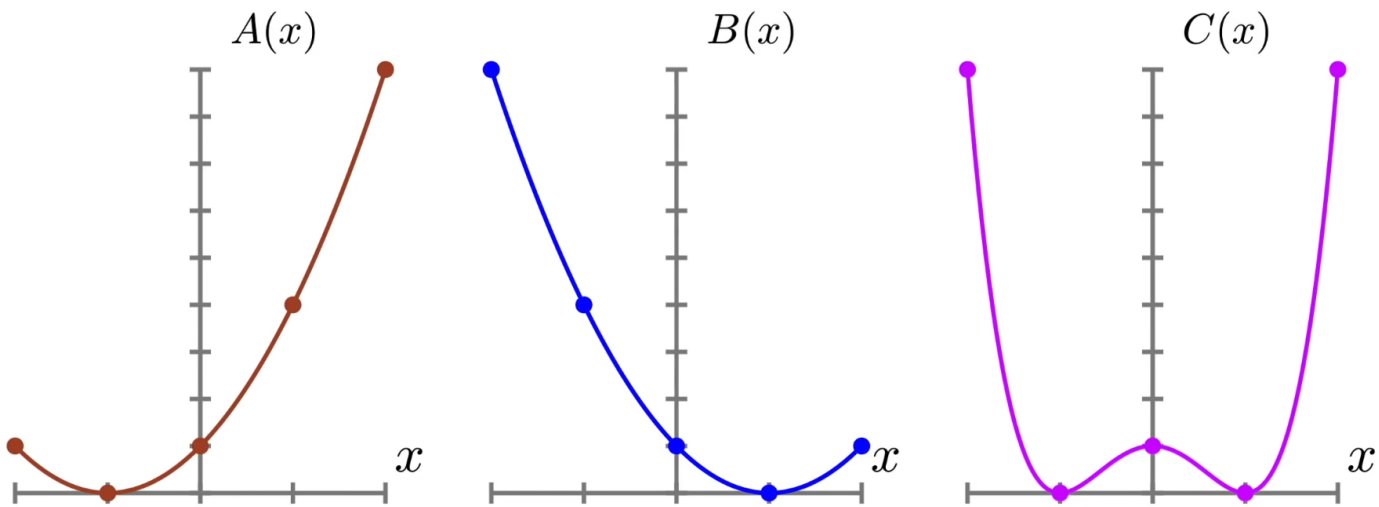
Sample and interpolate

- Sample $a(x)$ and $b(x)$ at $2n+1$ points
- Multiply each pair of samples
- Interpolate the points to get $c(x)$

$$A(x) = x^2 + 2x + 1 \qquad [(-2,1),(-1,0),(0,1),(1,4),(2,9)]$$

$$B(x) = x^2 - 2x + 1 \qquad [(-2,9),(-1,4),(0,1),(1,0),(2,1)]$$

$$C(x) = x^4 - 2x^2 + 1 \qquad [(-2,9),(-1,0),(0,1),(1,0),(2,9)]$$

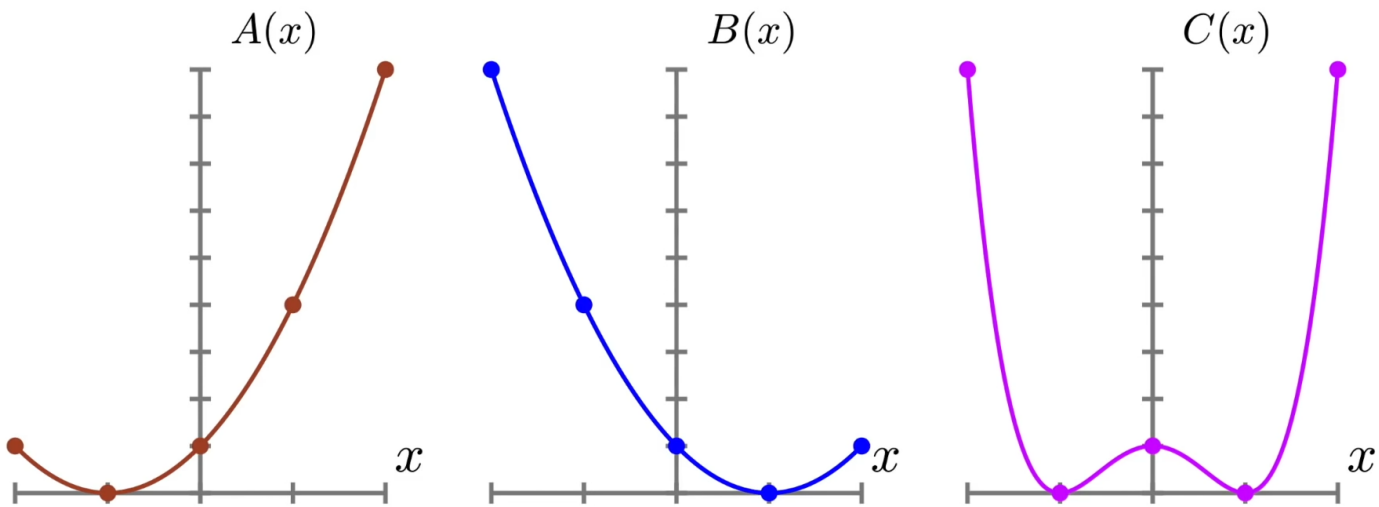


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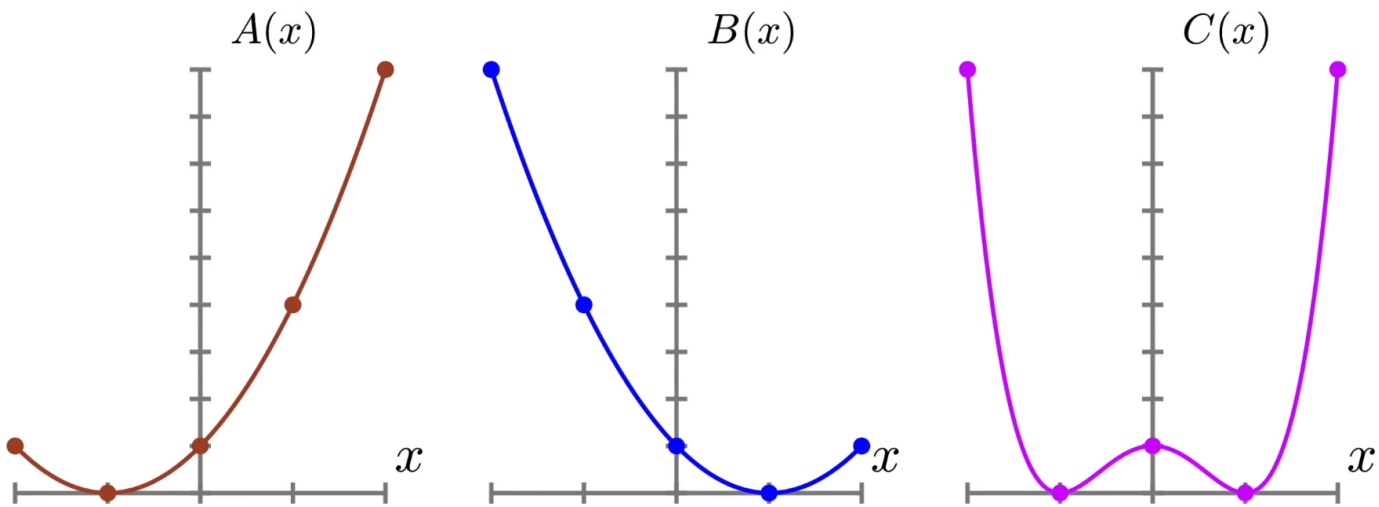


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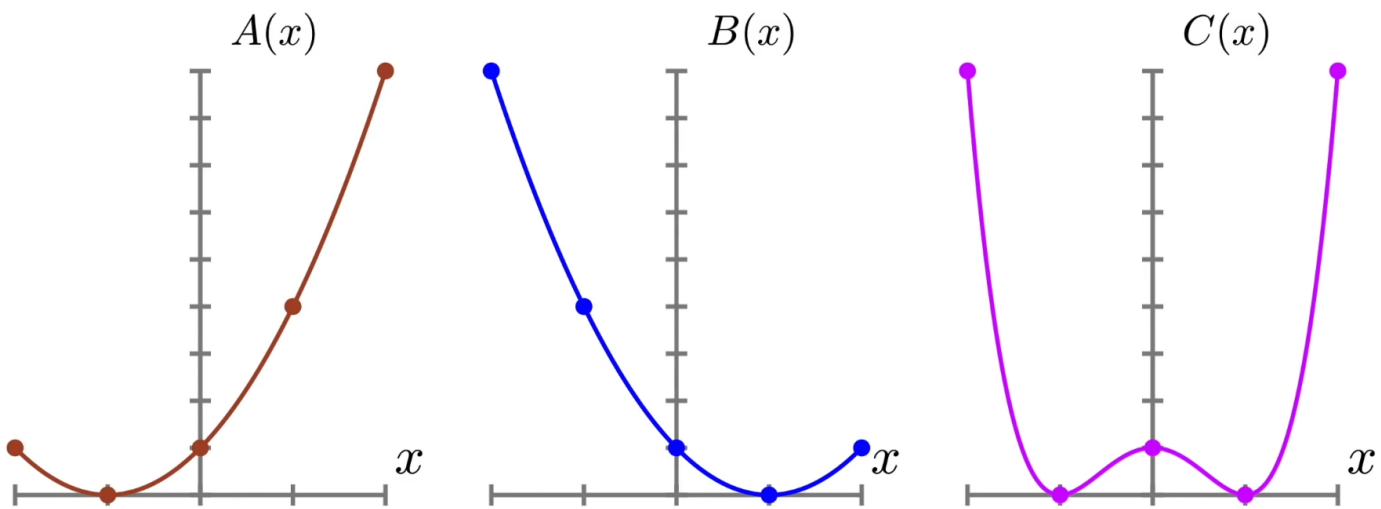


How can this be efficient?!

- Sampling: evaluate the polynomial at $2n+1$ points, $\mathcal{O}(n^2)$
- Pointwise multiplication: $\mathcal{O}(n)$
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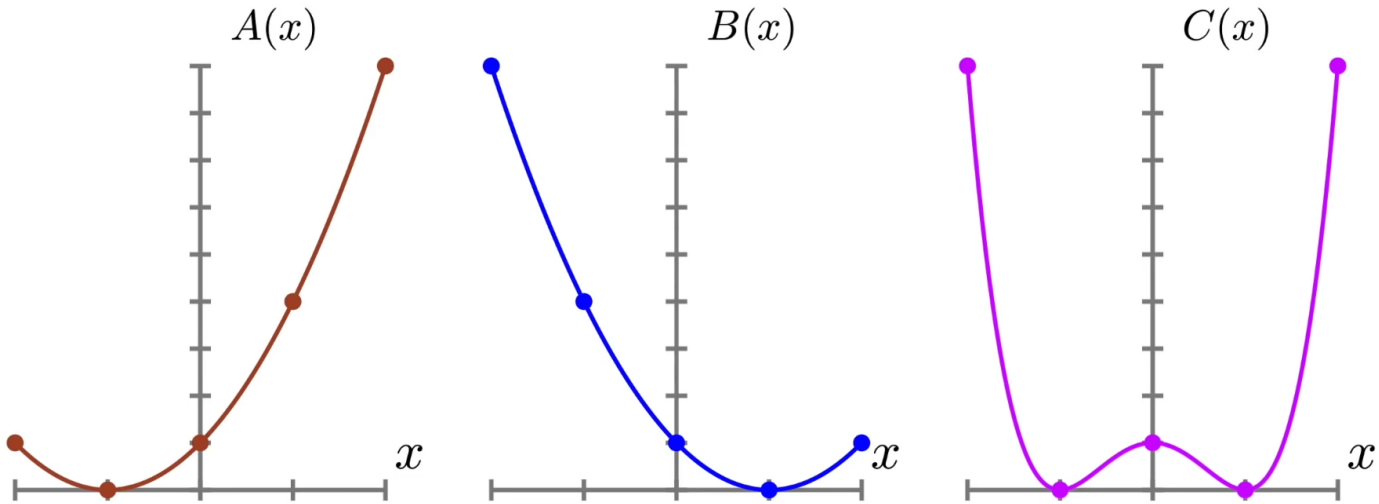


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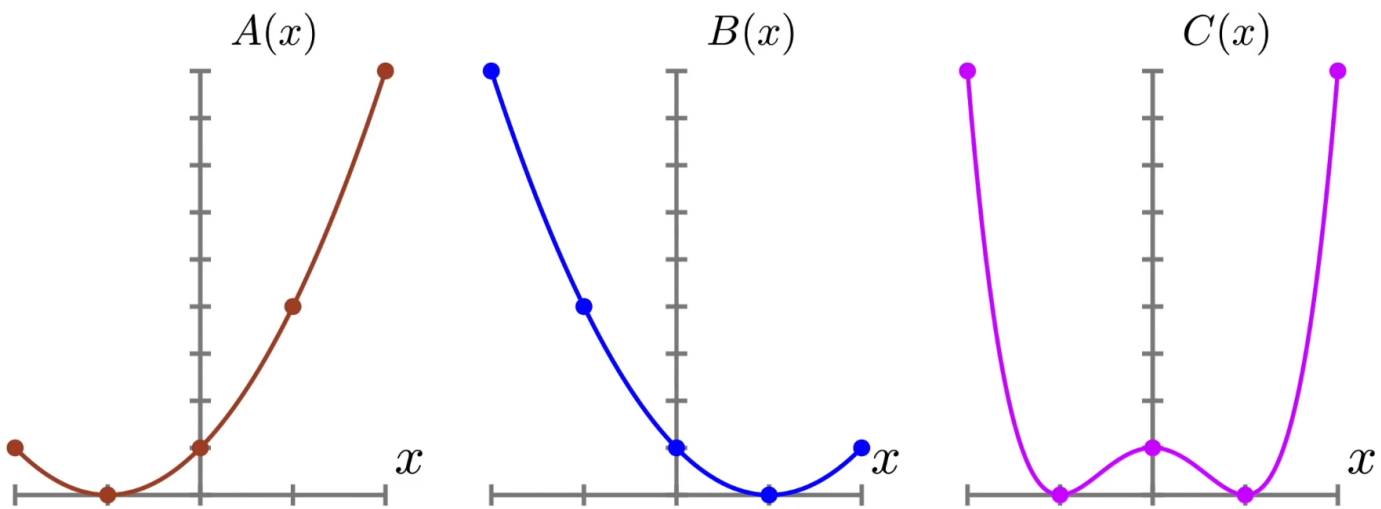


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The Discrete Fourier Transform (in $\mathbb{C}$)

Sampling

Evaluate the polynomial at the N-th roots of unity $\rightarrow$ take the DFT

$$\mathcal{F}(\omega_k) = \sum_{n=0}^{N} x_n e^{-\frac{2\pi i k}{N} n}$$

Interpolation

Interpolate the points $\rightarrow$ take the inverse DFT

$$\mathcal{F}^{-1}(X_k) = \frac{1}{N} \sum_{n=0}^{N} X_n e^{\frac{2\pi i k}{N} n}$$

Algorithms that calculate the DFT and its inverse are called Fast Fourier Transforms and can achieve $\mathcal{O}(n \log n)$ complexity.

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FFT in finite fields

- $a(x)$ and $b(x)$ are polynomials in $\mathbb{F}_2[x]$
- Each coefficient $a_i, b_i \in \mathbb{F}_2$ is a bit (0, 1)

Problem: Finite fields (of characteristic 2) do not admit roots of unity
Solution: use powers of a primitive element
Caveat(s):

- The degree must be a power of 2: $n = 2^\ell$
- Polynomials require a different representation

Case study: the Lin-Chung-Han FFT [LCH14]

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Polynomial evaluation (sampling)

We start from:

- $f(x) = g(X)$, a polynomial of degree 2^m in novelpoly basis
- $\mathbf{W}_m = \mathbb{F}_{2^m}$, the set of points to be evaluated

We want to get:

- $\{f(x)|x \in \mathbf{W}_m\}$, the evaluation of $f(x)$ at the points in $\mathbf{W}_m$

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The LCH FFT

Divide et impera

Recursively split:

- The polynomial: $f_0(X) + f_1(X)X_{2^k}$
- The set of points: $\mathbf{W}_{k-1}, (\mathbf{W}_k \setminus \mathbf{W}_{k-1}) = \{w + v_{k-1} | w \in \mathbf{W}_{k-1}\}$

Get 2 for the price of 1

Evaluate f at $\mathbf{W}_{k-1}$; recall that:

- Elements in novelpoly basis are combinations of vanishing polynomials
- Vanishing polynomials are linear: $s_i(x) + s_i(y) = s_i(x + y)$

Thus the evaluations at $(\mathbf{W}_k \setminus \mathbf{W}_{k-1})$ can be obtained starting from the evaluations at $\mathbf{W}_{k-1}$ with a *butterfly* unit

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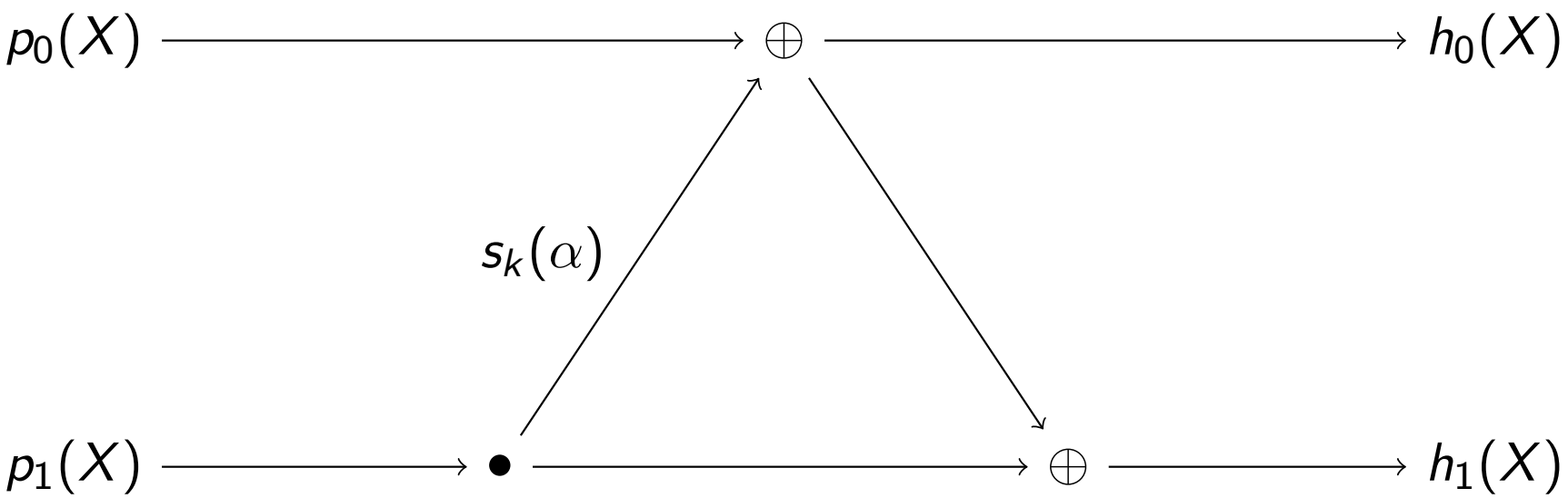
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Butterfly unit

Figure: Butterfly unit



Frobenius map

We can use the Frobenius map

$$\phi : \mathbb{F}_2 \to \mathbb{F}_2 \qquad x \mapsto x^2$$

to derive more values of a polynomial f of degree n from less evaluations.

Frobenius partition

We can build a set of points $\mathbf{S} \subset \mathbb{F}_{2^m}$ as:

$$\mathbf{S} := v_{l+m/2} + \mathbf{W}_l = \{v_{l+m/2} + x \mid x \in \mathbf{W}_l\}$$

which has size $n' = |\mathbf{S}| = |\mathbf{W}_l| = n/m$. By repeatedly applying ϕ to $\mathbf{S}$ we get

$$\mathbf{S} \cup \phi(\mathbf{S}) \cup \cdots \cup \phi^{m-1}(\mathbf{S}) = \mathbf{O}$$

with $|\mathbf{O}| = n$.

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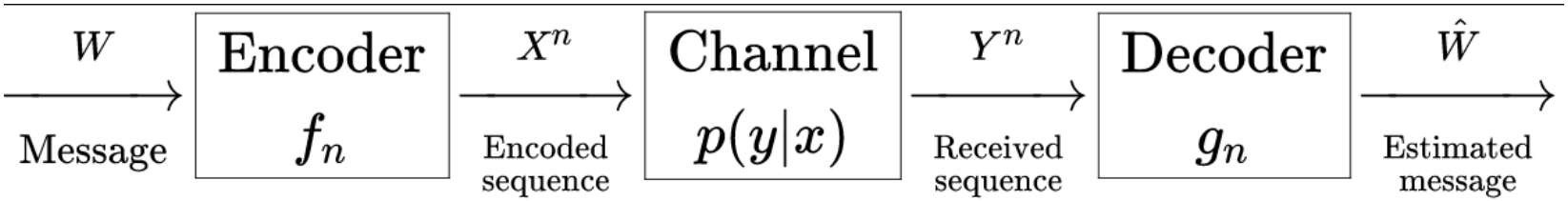
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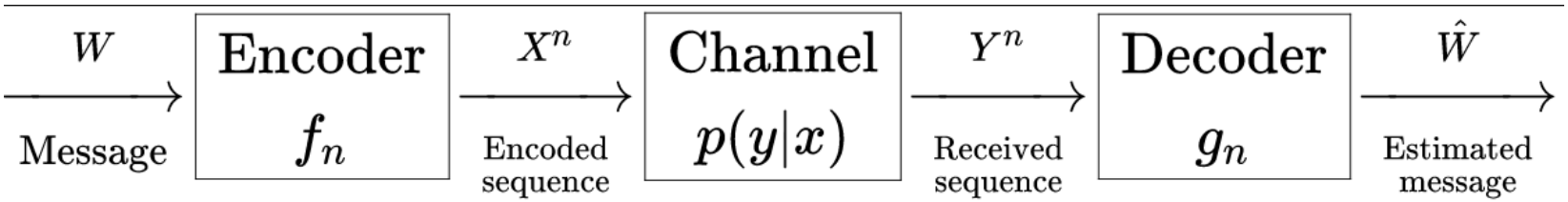


Problem: we want to send a message m over a noisy channel.

- The channel is noisy: it introduces errors in the message
- The receiver wants to recover the original message

Solution: send, along with the message, information that allows the receiver to correct the errors.

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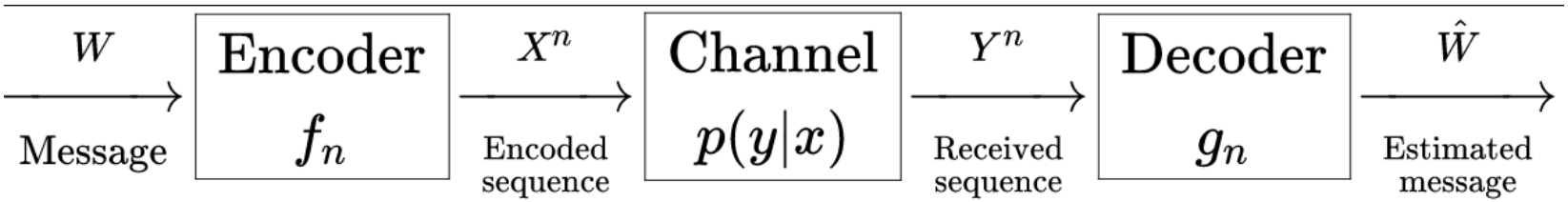


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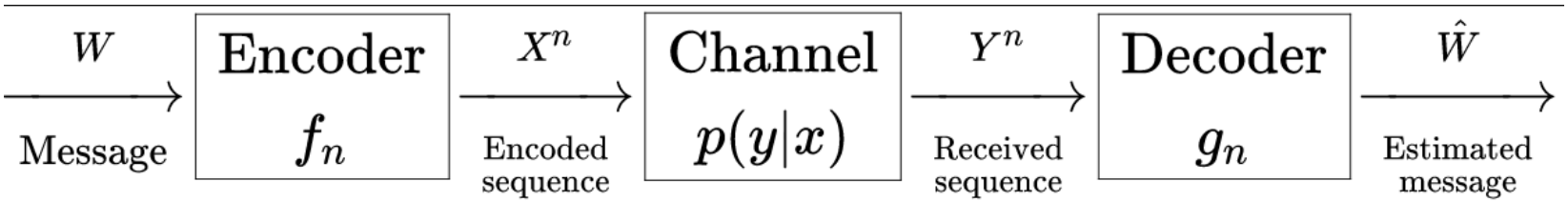


Problem: we want to send a message m over a noisy channel.

- The channel is noisy: it introduces errors in the message
- The receiver wants to recover the original message

Solution: send, along with the message, information that allows the receiver to correct the errors.

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Linear codes

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Let $\mathcal{V}$ be a vector space of dimension n over $\mathbb{F}_2$. A linear code $\mathcal{C} = [n, k]$ is a subspace of $\mathcal{V}$ of dimension k.

- Messages are k bits long
- We add $n - k$ bits to detect and correct errors (up to δ)

Generator Matrix

The matrix $\mathbf{G} \in \mathbb{F}_2^{k \times n}$ is the generator matrix of $\mathcal{C}$ if

$$\mathcal{C} = \{\mathbf{mG} \mid \mathbf{m} \in \mathbb{F}_2^k\}$$

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Syndrome

Parity Check Matrix

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$$\mathcal{C} = \{\mathbf{v} \in \mathbb{F}_2^n \mid \mathbf{H}\mathbf{v}^\top = 0\}$$

The syndrome of a word $\mathbf{v} \in \mathcal{C}$ is defined as:

$$\mathbf{S}(\mathbf{v}) = \mathbf{H}\mathbf{v}^\top$$

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Syndrome Decoding

Given an encoded message, find the closest message w.r.t. a certain metric (*i.e.* Hamming weight, $\mathtt{wt}(\cdot)$).

How hard can this be?

- Take a word $\mathbf{x}$ with $\mathtt{wt}(w)$
- Calculate the syndrome $\mathbf{y}^\top = \mathbf{H}\mathbf{x}^\top$
- Given $\mathbf{H}$ and $\mathbf{y}^\top$, retrieve $\mathbf{x}$

This problem is **NP-complete** [BMvT78].

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Hamming Quasi-Cyclic

- IND-CPA Public Key Encryption
- IND-CCA2 Key Encapsulation Mechanism
- Based on a variant of the syndrome decoding problem
- Employs quasi-cyclic codes to shorten key sizes

Codes

HQC employs two codes:

- a decodable $\mathcal{C} = [n, k]$ generated by $\mathbf{G} \in \mathbb{F}_2^{k \times n}$ (public)
- a random $[2n, n]$ code with parity-check matrix $\mathbf{H} = (\mathbf{I}_n | \mathbf{circ}(\mathbf{h}))$

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Public Key Encryption

Setup

Generate the parameters $(n, k, \delta, w, w_r, w_e)$ for the corresponding security level λ

Key Generation

- Generate $\mathbf{h} \xleftarrow{\$} \mathbb{F}_2^n$
- Generate $(\mathbf{x}, \mathbf{y}) \xleftarrow{\$} \mathbb{F}_2^n$, with $\mathtt{wt}(\mathbf{x}) = \mathtt{wt}(\mathbf{y}) = w$
- Calculate $\mathbf{s} = \mathbf{x} + \mathbf{h}\mathbf{y}$

$$\mathcal{K}_{pub} = (\mathbf{h}, \mathbf{s}) \qquad \mathcal{K}_{pri} = (\mathbf{x}, \mathbf{y})$$

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To encrypt a message $\mathbf{m} \in \mathbb{F}_2^k$:

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- Calculate $\mathbf{u} = \mathbf{r_1} + \mathbf{h}\mathbf{r_2}$
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$$\mathrm{ctx} = (\mathbf{u}, \mathbf{v})$$

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Decryption

From the ctx $= (\mathbf{u}, \mathbf{v})$ and the private key $\mathcal{K}_{pri} = (\mathbf{x}, \mathbf{y})$:

$$\begin{aligned}\mathbf{v} - \mathbf{u} \cdot \mathbf{y} &= (\mathbf{mG} + \mathbf{s} \cdot \mathbf{r}_2 + \mathbf{e}) - (\mathbf{r}_1 + \mathbf{h} \cdot \mathbf{r}_2)\mathbf{y} = \mathbf{mG} + \mathbf{x} \cdot \mathbf{r}_2 - \mathbf{r}_1 \cdot \mathbf{y} + \mathbf{e} \\ &= \mathbf{mG} + (\mathbf{x} + \mathbf{h} \cdot \mathbf{y})\mathbf{r}_2 + \mathbf{e} - (\mathbf{r}_1 \cdot \mathbf{y} + \mathbf{h} \cdot \mathbf{r}_2 \cdot \mathbf{y}) \\ &= \mathbf{mG} + \mathbf{x} \cdot \mathbf{r}_2 + \mathbf{h} \cdot \mathbf{y} \cdot \mathbf{r}_2 + \mathbf{e} - \mathbf{r}_1 \cdot \mathbf{y} - \mathbf{h} \cdot \mathbf{r}_2 \cdot \mathbf{y} \\ &= \mathbf{mG} + \mathbf{x}_2 - \mathbf{r}_1 \cdot \mathbf{y} + \mathbf{e}\end{aligned}$$

So we have $\mathbf{m} = \mathcal{C}.\text{Decode}(\mathbf{v} - \mathbf{u} \cdot \mathbf{y})$ whenever

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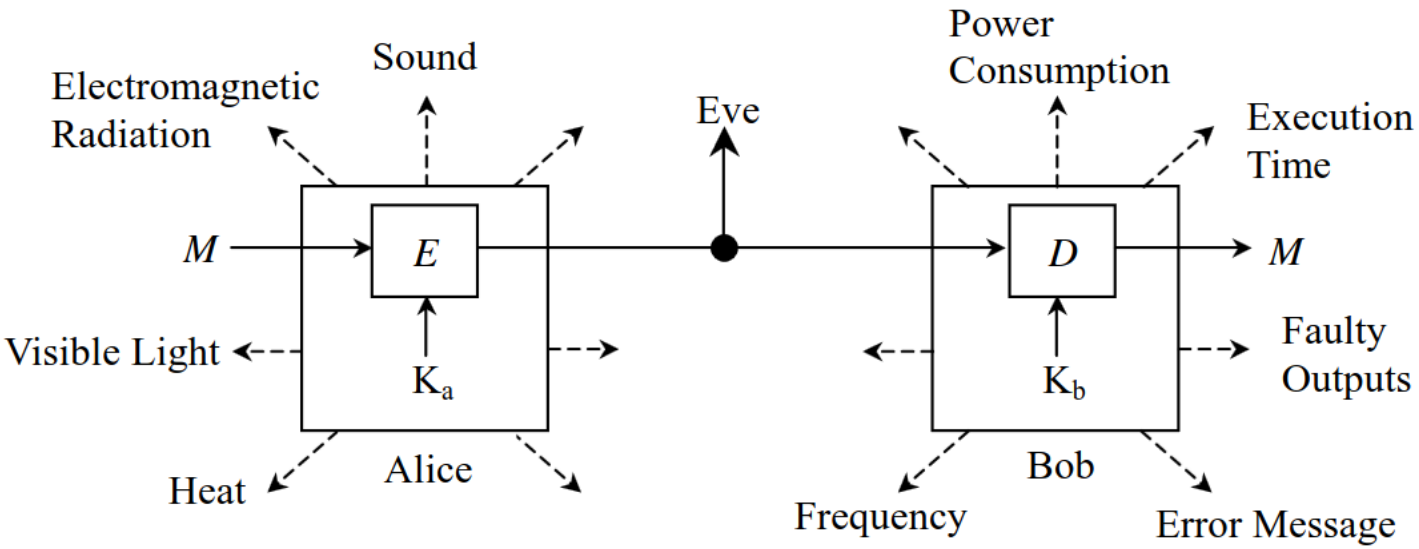
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Side channel attacks

Devices running cryptosystems may expose secrets through *side channels*
Observable aspects of the system from which information can be extracted



Implementation

Parameters

Our target: `hqc-128` on ARM Cortex-M4

Multiplication parameters

- Maximum polynomial degree: $n = 17669$
- Hamming weight of sparse polynomials: $\omega = 75$

Target board

STM32L476RG

- 32-bit CPU
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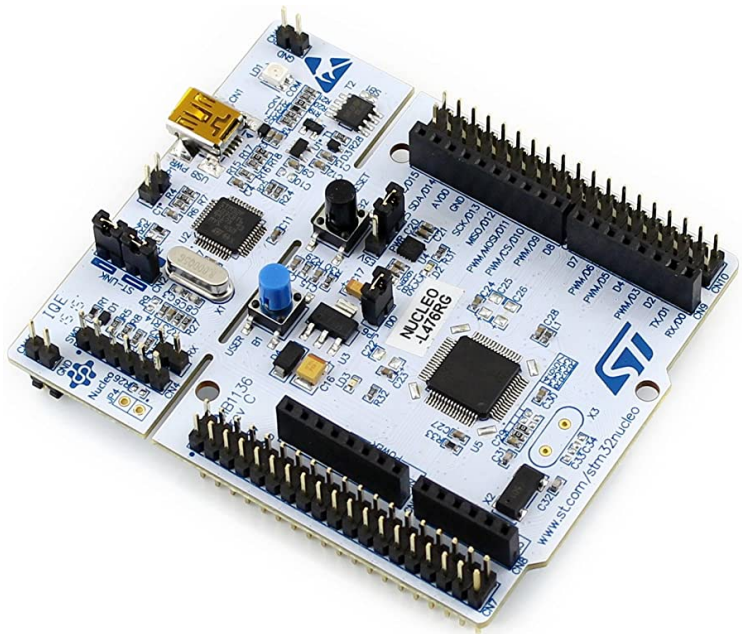
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Masking HQC

HQC operations are linear $\rightarrow$ boolean masking of the entire scheme

$$a \mapsto [\![a]\!] = (a_1, \cdots, a_d) \text{ such that } \bigoplus_{i=1}^{d} a_i = a$$

Our implementation

More shares $\rightarrow$ slower performance, higher security

We chose $d = 2$ shares:

$$[\![a]\!] = (a \oplus x, x)$$

- a is the secret value
- x is a random value (from the TRNG)

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Keygen

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Mask x and y, perform the same operations of the unmasked version

Function KEYGEN()

$h \xleftarrow{\$} \mathcal{R}$;

$(x, y) \xleftarrow{\$} \mathcal{R}_\omega \times \mathcal{R}_\omega$;

$[\![y]\!] \leftarrow (y + \sum \mathtt{rnd}_i)$;

$[\![s]\!] \leftarrow h[\![y]\!]$;

$[\![s]\!] \leftarrow x + [\![s]\!]$;

$\mathcal{K}_{pri} \leftarrow (x, y)$;

$\mathcal{K}_{pub} \leftarrow (\text{COLLAPSE}([\![h]\!]), s)$;

return $(\mathcal{K}_{pri}, \mathcal{K}_{pub})$

Encryption

Mask e, r_1 and r_2, perform the same operations of the unmasked version

Function ENCRYPT($\mathcal{K}_{pub}, m$)

$e \xleftarrow{\$} \mathcal{R}_{\omega_e}$;

$(r_1, r_2) \xleftarrow{\$} \mathcal{R}_{\omega_r} \times \mathcal{R}_{\omega_r}$;

$[\![r_1]\!] \leftarrow (r_1 + \sum \mathtt{rnd}_i)$;

$[\![r_2]\!] \leftarrow (r_2 + \sum \mathtt{rnd}_i)$;

$[\![u]\!] \leftarrow [\![r_1]\!] + h[\![r_2]\!]$;

$[\![m]\!] \leftarrow (m + \sum \mathtt{rnd}_i)$;

$[\![v]\!] \leftarrow [\![m]\!]\mathbf{G} + s[\![r_2]\!] + e$;

return (COLLAPSE($[\![u]\!]$), COLLAPSE($[\![v]\!]$))

Decryption

Function DECRYPT($\mathcal{K}_{pri}, c$)

$[\![y]\!] \leftarrow (y + \sum \mathtt{rnd}_i)$;

$[\![uy]\!] \leftarrow u[\![y]\!]$;

$[\![\bar{m}]\!] \xleftarrow{\$} (\{0,1\}^k)^d$;

$[\![\bar{c}]\!] \leftarrow [\![\bar{m}]\!]\mathbf{G}$;

$a \leftarrow$ COLLAPSE$((v\ , [\![\bar{c}]\!]))$;

$m \leftarrow$ DECODE(COLLAPSE$(a, [\![uy]\!])$);

return COLLAPSE$(m\ , [\![\bar{m}]\!])$;

Generate a random message $[\![\bar{m}]\!]$

Encode it $[\![\bar{c}]\!] = [\![\bar{m}]\!]\mathbf{G}$

Add the first half of the ciphertext

Collapse it together with the second half of the ciphertext

Decode the result

Remove the random message

Linearity of the RMRS code

$$\text{ENCODE}(m_1 + m_2) = \text{ENCODE}(m_1) + \text{ENCODE}(m_2)$$

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$[\![uy]\!] \leftarrow u[\![y]\!]$;
$[\![\bar{m}]\!] \xleftarrow{\$} (\{0,1\}^k)^d$;
$[\![\bar{c}]\!] \leftarrow [\![\bar{m}]\!]\mathbf{G}$;
$a \leftarrow$ COLLAPSE$((v\ , [\![\bar{c}]\!]))$;
$m \leftarrow$
DECODE(COLLAPSE$(a, [\![uy]\!])$);
return COLLAPSE$(m\ , [\![\bar{m}]\!])$;

Generate a random message $[\![\bar{m}]\!]$
Encode it $[\![\bar{c}]\!] = [\![\bar{m}]\!]\mathbf{G}$
Add the first half of the ciphertext
Collapse it together with the second half of the ciphertext
Decode the result
Remove the random message

Linearity of the RMRS code

$$\text{ENCODE}(m_1 + m_2) = \text{ENCODE}(m_1) + \text{ENCODE}(m_2)$$

Experimental results

Measuring time

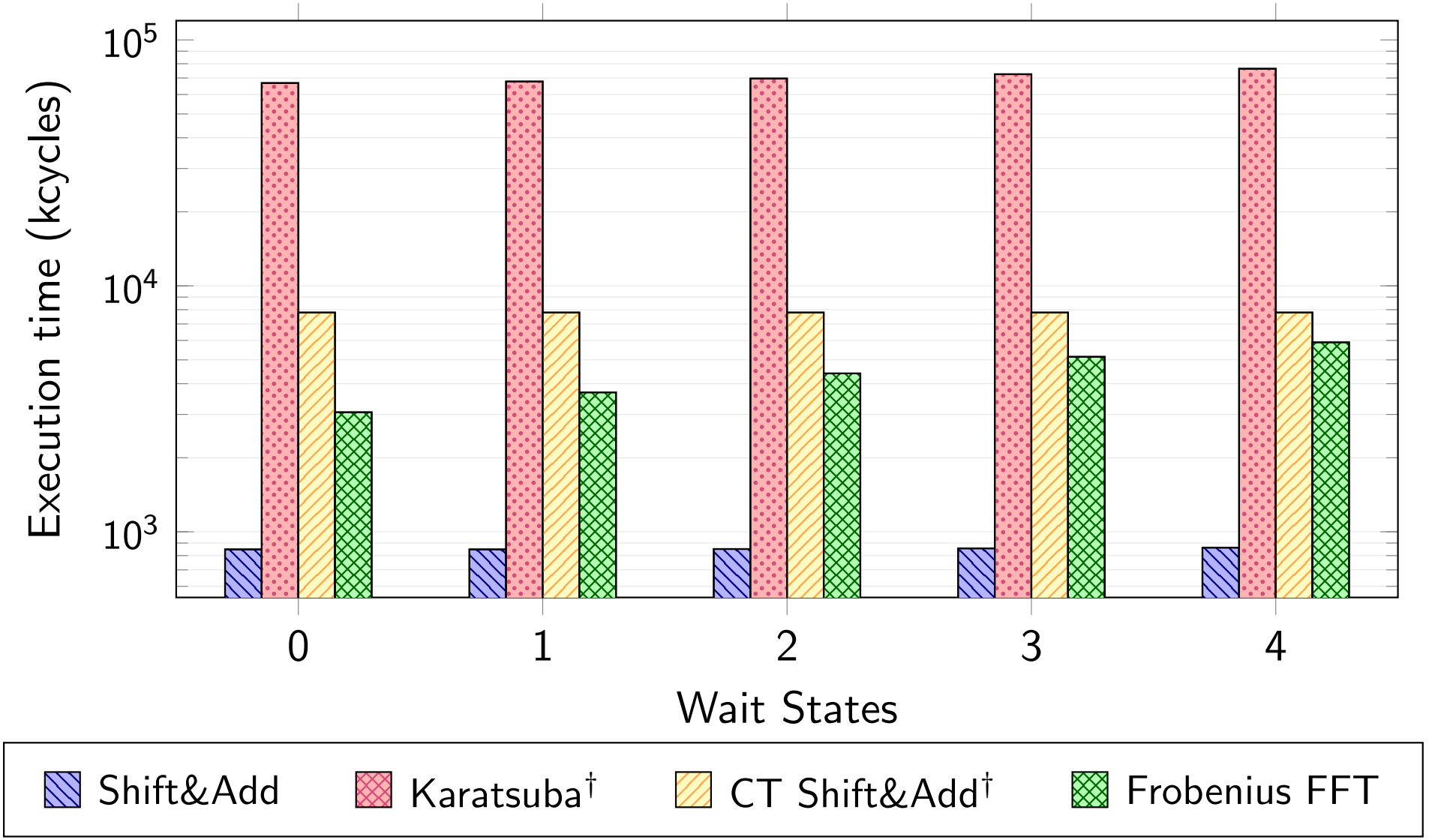
Performance evaluation metric: amount of clock cycles

- `DWT_CYCCNT` register for Cortex-M4
- `rtdsc` instruction for x86

Measurements are averaged over 1000 runs.
Repeat measurements while varying the amount of memory wait states.

- 16, 32, 48, 64 and 80 MHz on ARM
- 2GHz + no TurboBoost on x86

Timing results



Test Vector Leakage Assessment

Statistical t-test

Determine how much the means of two populations differ by evaluating

$$t = \frac{\mu_0 - \mu_1}{\sqrt{\frac{s_0^2}{n_0} + \frac{s_1^2}{n_1}}}$$

A t-value $|t| < 4.5$ implies that the populations have the same mean with 99.999% confidence.

Populations to be measured (1000 rounds each)

- fixed inputs
- random inputs

Randomness is always generated by the onboard TRNG.

Constant-time: multiplication

Table: t-values for the multiplication algorithms

	Shift&Add	**Karatsuba**	**CT Shift&Add**	**Frobenius FFT**
`ARM`	55.748	0.000	0.000	0.000
`x86`	20.936	3.644	3.371	1.232

All algorithms except plain S&A are constant-time, we choose

- Frobenius FFT on ARM
- Karatsuba (with AVX) on x86

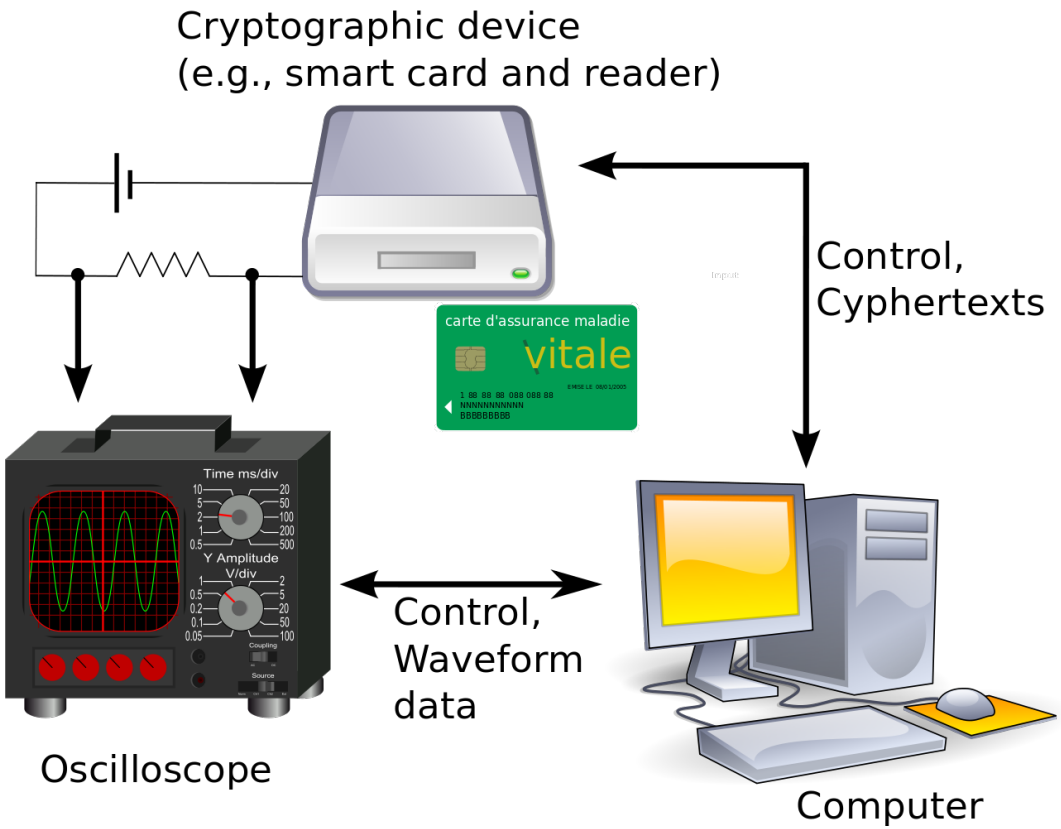
Constant-time: PKE

Round	Version	Mul	CT?	Encrypt x86	Encrypt ARM	Decrypt x86	Decrypt ARM
3	`reference`	Shift&Add	○	10.65	29.19	5.59	39.23
	`sparsemul`	CT Shift&Add†	•	9.38	23.12	7.59	49.56
4	`reference`	Karatsuba†	•	1.27	1.04	0.97	0.30
	`sparsemul`	CT Shift&Add†	•	0.29	0.07	1.51	1.64
	`fastmul`	Frobenius FFT	•	3.36	0.56	1.70	1.99

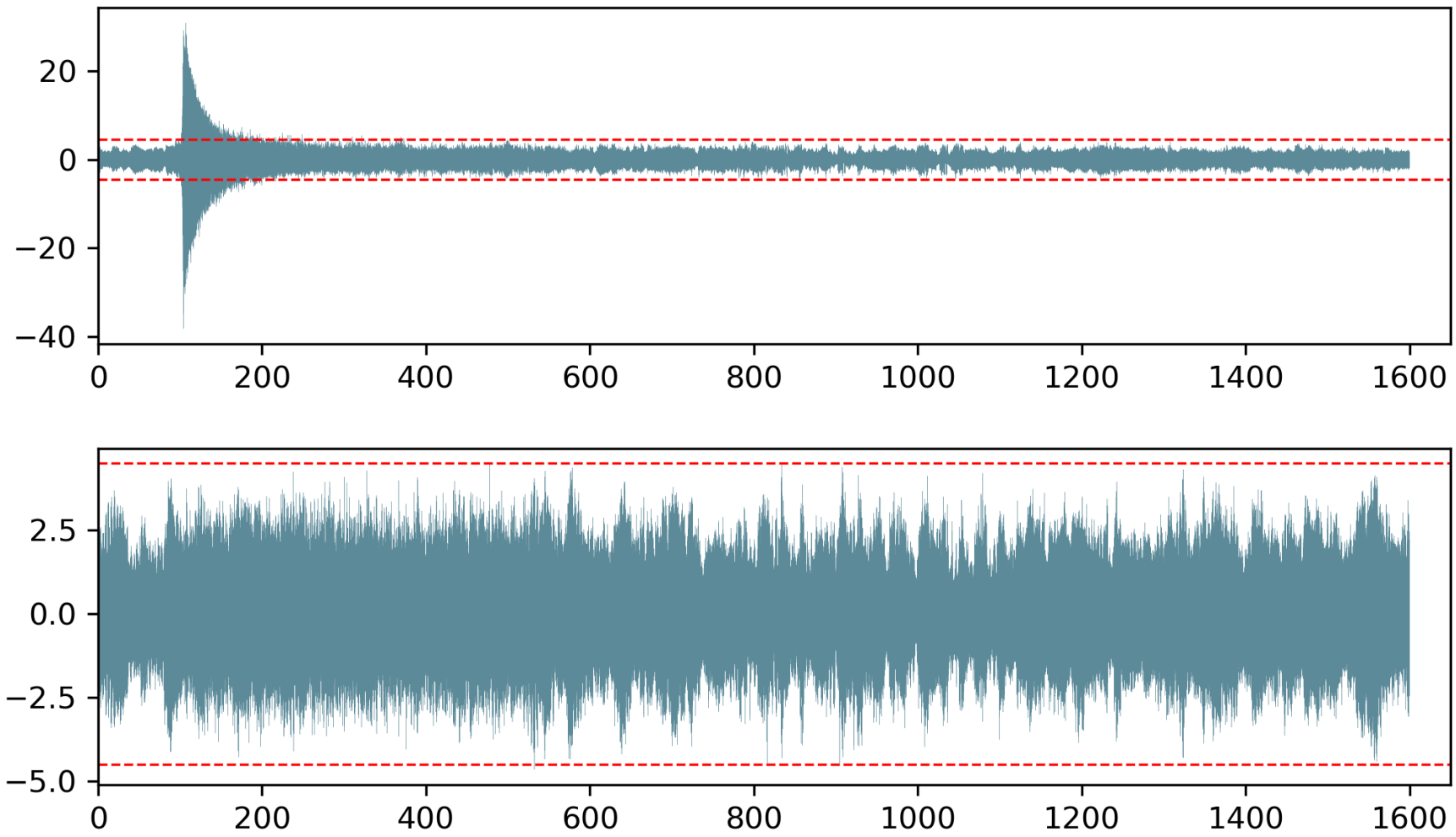
Power measurements

Setup

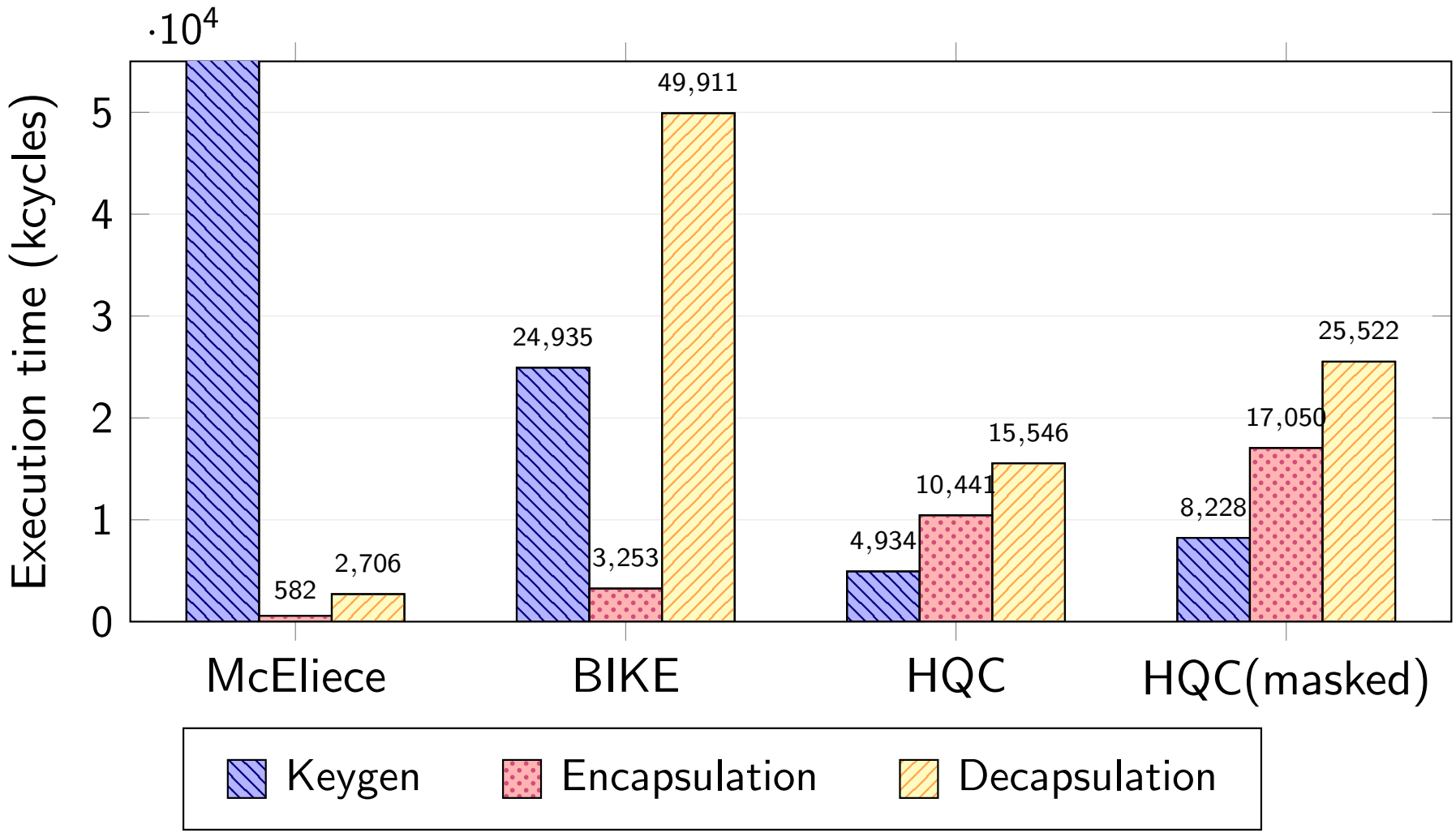
Measure the EM radiation emitted by the board package (non invasive)



EM t-values: decoder



Execution time



Conclusions

Results

Our implementation

- Fast and secure multiplication algorithms
- Constant-time PKE and KEM
- Efficient masking scheme for the PKE

Future works

Extension of the masking scheme to the KEM

Thanks for your attention!

Questions?

Appendix

Context

Cryptography 101

Kerchoff's Principle

Everything about a cryptosystem is publicly known, **except for the key**

Symmetric encryption

Key: a n-bit string

Asymmetric encryption

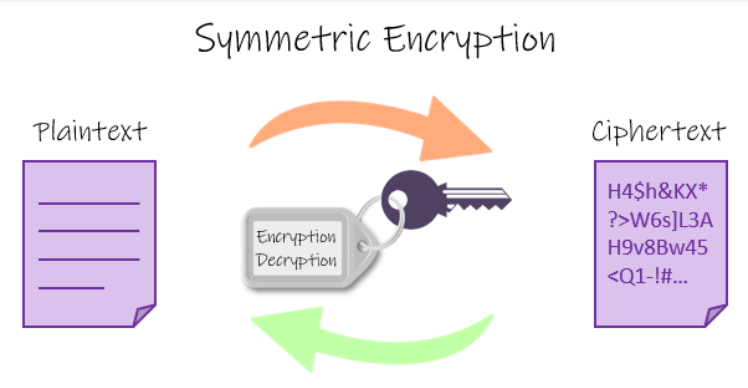
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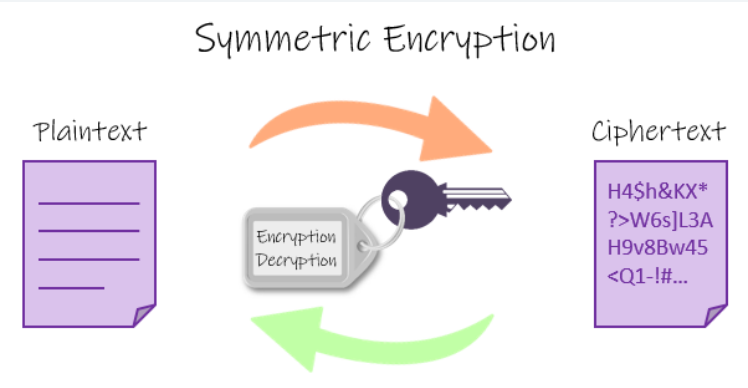
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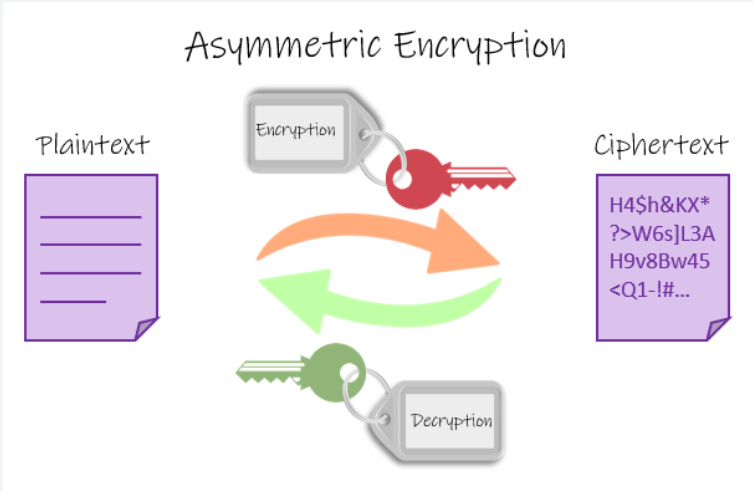
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Cryptanalytic attacks

What are the best attacks on modern cryptosystems?

On Classical Computers

- Symmetric: exhaustive search, $\mathcal{O}(2^n)$
- Asymmetric: General Number Field Sieve, $\approx \mathcal{O}(2^{n-\epsilon})$

On Quantum Computers

- Symmetric: Grover's algorithm, $\mathcal{O}(\sqrt{2^n})$
- Asymmetric: Shor's algorithm, $\approx \mathcal{O}((\log n)^2)$

Shor's algorithm can break asymmetric cryptosystems in polynomial time!

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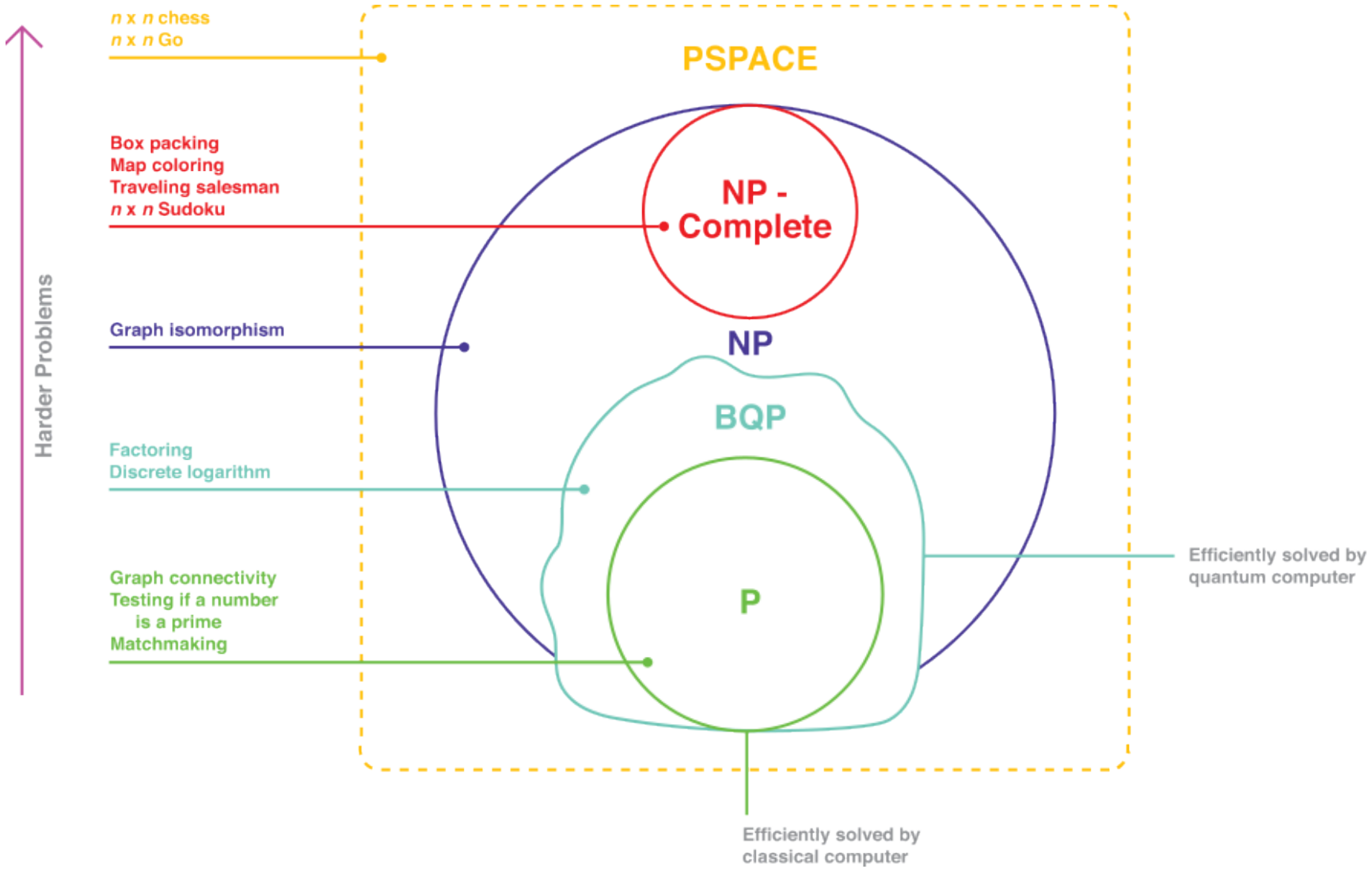
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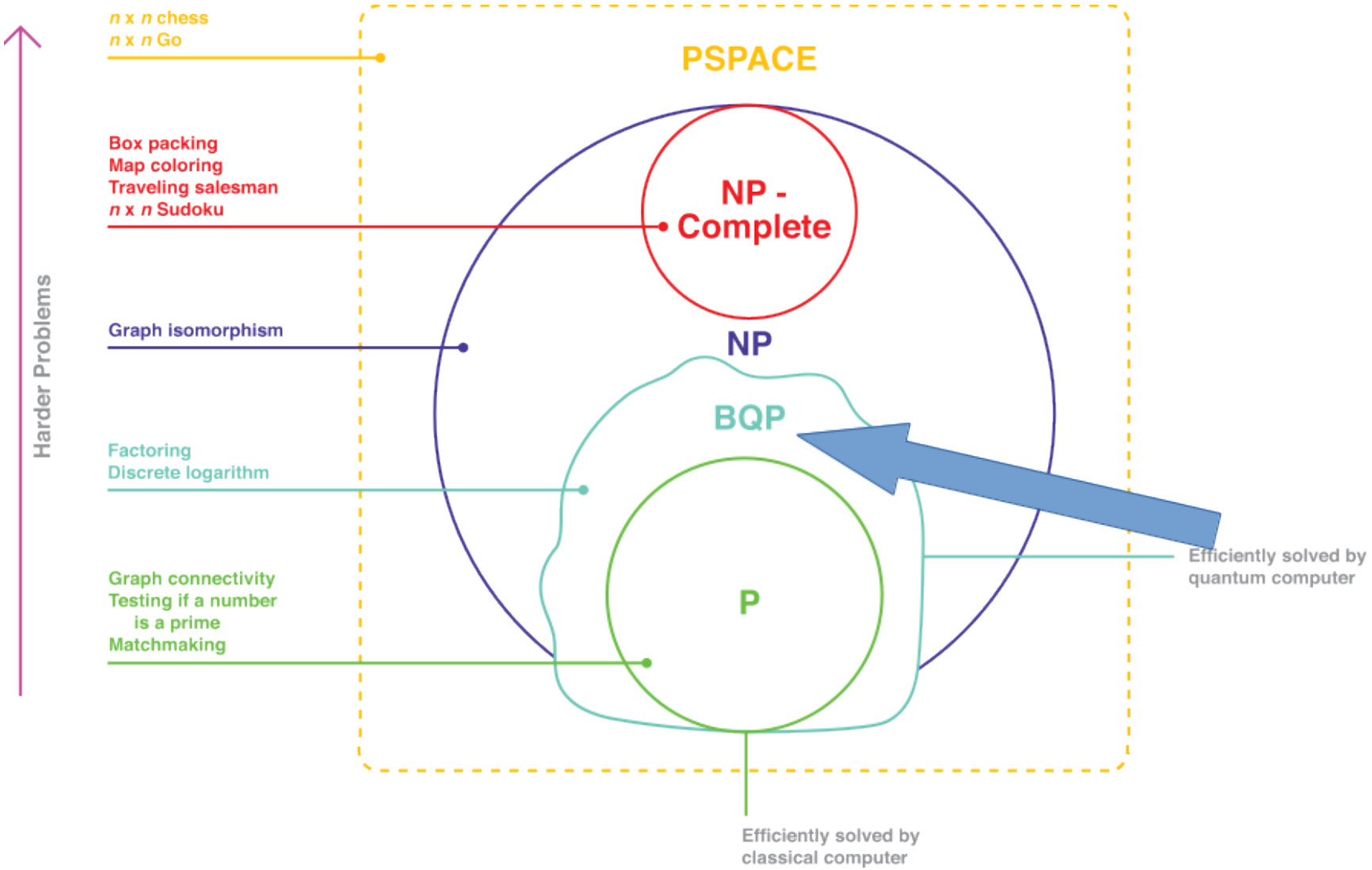
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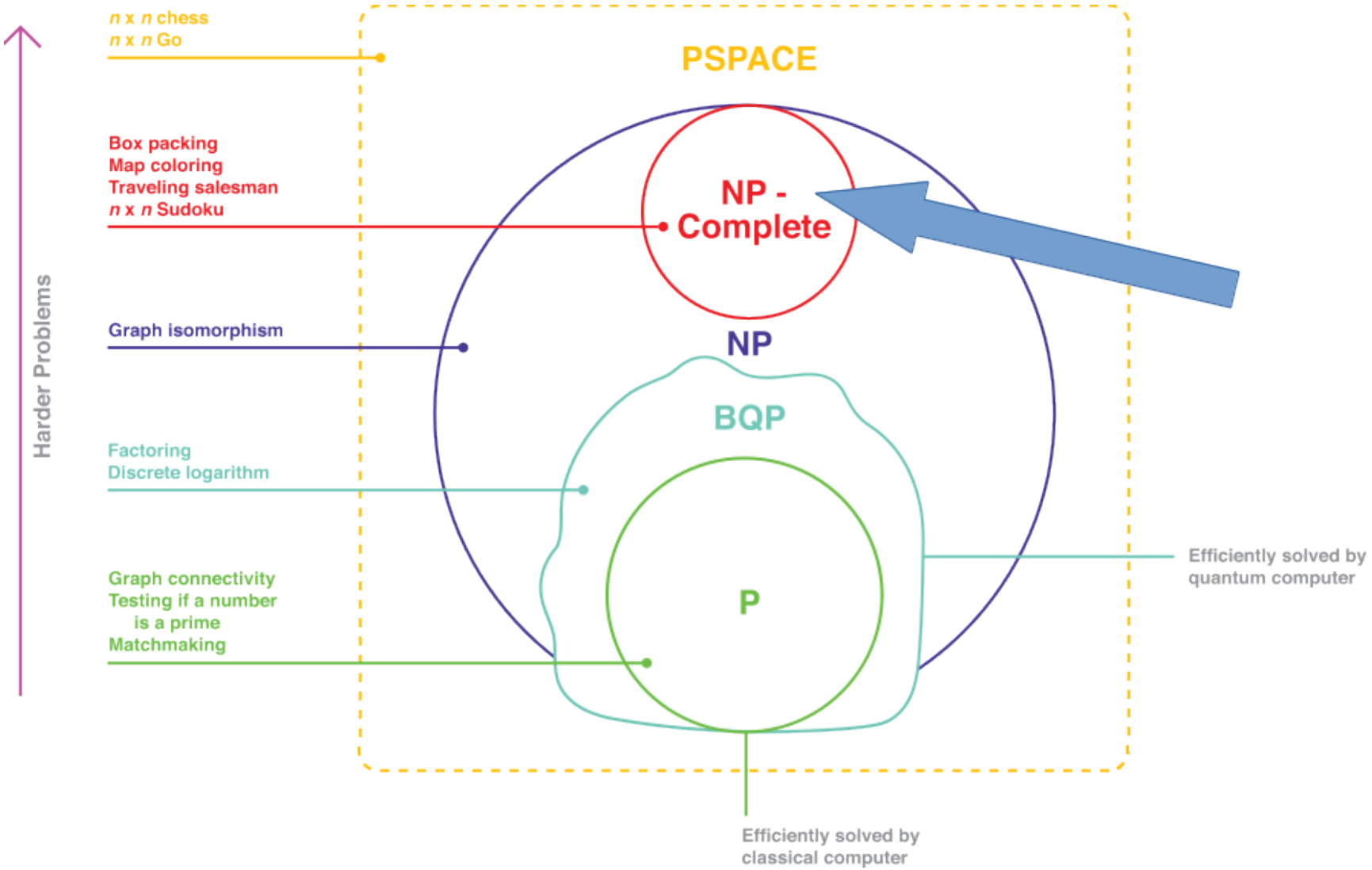
The Computational Complexity Zoo



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NIST Post-Quantum Cryptography Standardization

Timeline

- 26/02/16: Announcement of NIST's call for submissions
- 21/12/17: Announcement of the first round candidates (69)
- 30/01/19: Announcement of the second round candidates (26)
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Standardization candidates

Third Round winner

- CRYSTALS-Kyber: lattice-based

NIST decided to standardize at least another PQC scheme, in case lattice-based crypto fails

Fourth Round candidates

- ~~**S**upersingular **I**sogeny **K**ey **E**ncapsulation~~
- **B**it Fl**i**pping **K**ey **E**ncapsulation [ABB+22]
- **H**amming **Q**uasi-**C**yclic [MAB+18]
- Classic McEliece [BCC+22]

BIKE, Classic McEliece and HQC are all code-based cryptosystems.

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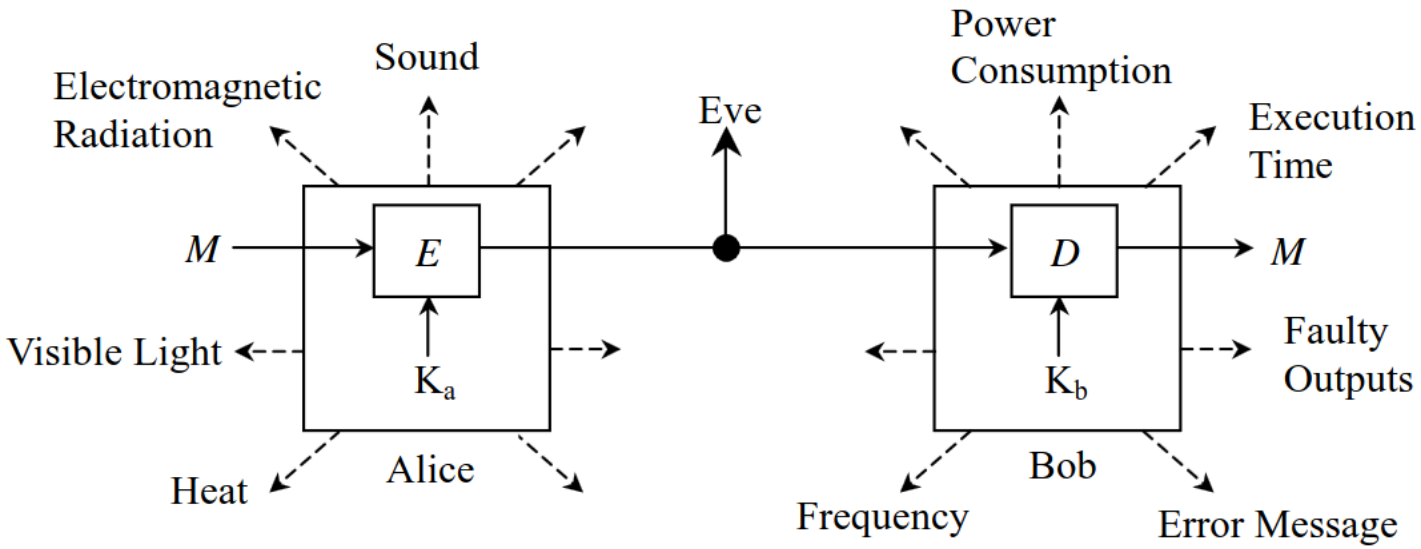
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Side channel attacks

Side channel attacks

Devices running cryptosystems may expose secrets through *side channels*
Observable aspects of the system from which information can be extracted



Timing attacks

Execution time may vary based on the inputs

- Loops iterating over the input
- Branches based on the input

Countermeasures

Don't use sensitive inputs in branching instructions

- Use constant time algorithms

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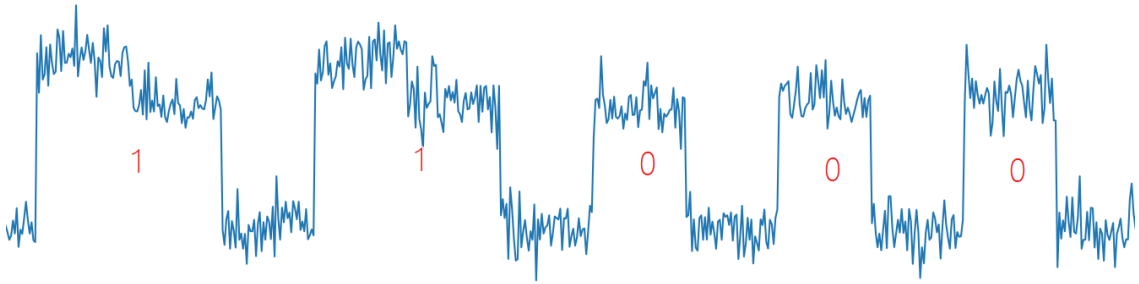
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Power attacks

Different operations require different amounts of power
Applies to both power and EM attacks



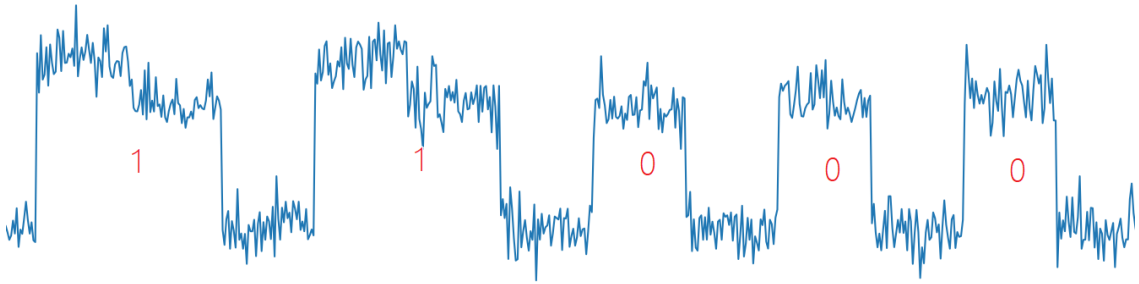
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Mask sensitive data with random values

- Makes the power consumption fluctuate randomly

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Basis conversion

Cantor basis

Vector spaces and bases

$\mathbb{F}_2^m$: vector space of m-bit strings over $\mathbb{F}_2$, with m power of 2
Elements of $\mathbb{F}_2^m$ are *vectors*, linear combinations of the *basis vectors*
Basis vectors: $\mathbf{v} = (v_1, \cdots, v_m)$

Canonical basis

$$\mathbf{v} = (1, \alpha, \alpha^2, \cdots, \alpha^{m-1})$$

Cantor basis

$$\mathbf{v} = (v_1, \cdots, v_m)$$

$$v_1 = 1 \qquad v_i^2 + v_i = v_{i-1}$$

Subspaces and vanishing polynomials

$$v_1 = 1 \qquad v_i^2 + v_i = v_{i-1}$$

Subspaces induced by Cantor basis

$$\mathbf{W}_0 := \{0\}, \qquad \mathbf{W}_i := span\{v_0, \cdots, v_{i-1}\}$$

with $\mathbf{W}_0 \subset \mathbf{W}_1 \subset \cdots \subset \mathbf{W}_m = \mathbb{F}_2^m$ and $\dim(\mathbf{W}_i) = i$.

Vanishing polynomials

$$s_i(x) = \prod_{a \in \mathbf{W}_i} (x - a)$$

with $\deg(s_i(x)) = |\mathbf{W}_i| = 2^i$.

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Novelpoly basis

Properties of the vanishing polynomials

- linearity: $s_i(x) + s_i(y) = s_i(x + y)$
- recursivity: $s_{i+j}(x) = s_i(s_j(x))$, $s_i(v_{i+j}) = v_j$
- $s_i(v_i) = 1$ for v_i element of the Cantor basis ([CCK+17])
- all v_i are roots of $s_{i+1}(x)$

Novelpoly basis

Cantor basis $\rightarrow$ Subspaces $\rightarrow$ Vanishing polynomials $\rightarrow$ Novelpoly basis

$$X_k(x) = \prod (s_i(x))^{b_i} \qquad \text{with } k = \sum_{i=0}^{\log_2 n} b_i 2^i$$

with $\deg(X_k(x)) = k$.

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FFT Butterflies

LCH FFT - example

Consider a polynomial f in novelpoly basis of degree greater or equal than $\deg(s_k(x)) = 2^k$ and split it in half:

$$f(x) = p_0(x) + s_k(x)p_1(x)$$

Evaluate $p_0(x)$ and $p_1(x)$ at $\mathbf{W}_{k-1} = \{\alpha \mid \alpha \in \mathbf{W}_{k-1}\}$:

$$p_0(\alpha) + s_k(\alpha)p_1(\alpha)$$

Evaluate $p_0(x)$ and $p_1(x)$ at $(\mathbf{W}_k \setminus \mathbf{W}_{k-1}) = \{\alpha + v_k \mid \alpha \in \mathbf{W}_{k-1}\}$

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$p_0(x)$ and $p_1(x)$ are combinations of vanishing polynomials $s_i(x)$ with $i < k$:

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$$p_0(\alpha + v_k) = p_0(\alpha), \qquad p_1(\alpha + v_k) = p_1(\alpha)$$

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Figure: Butterfly unit

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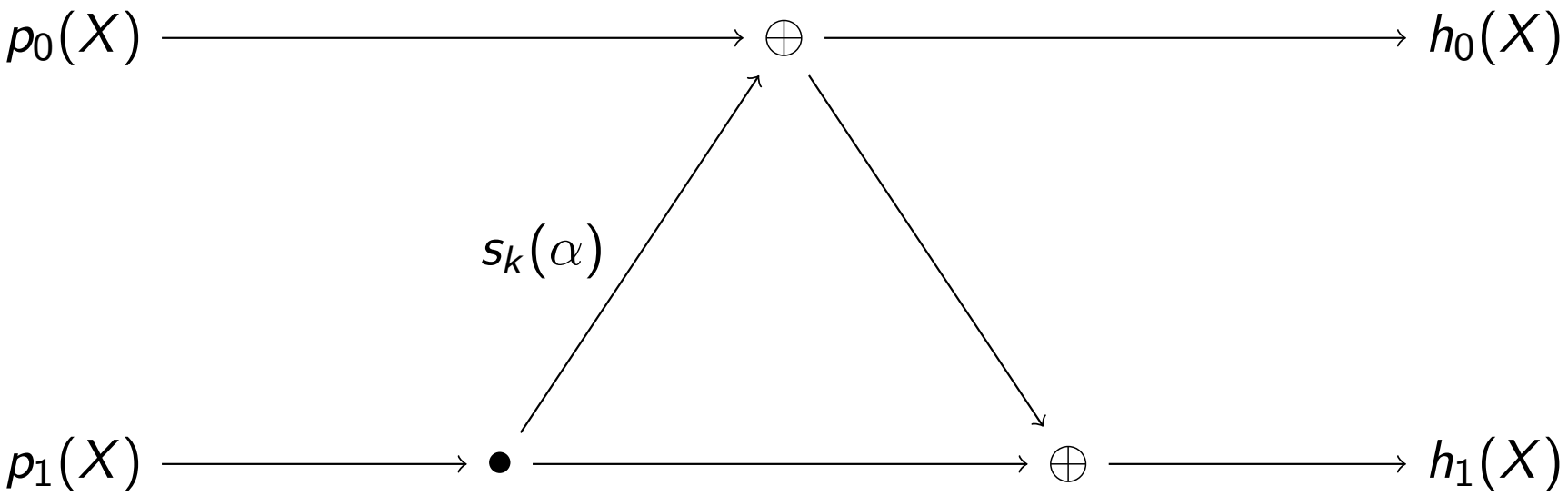
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Fujisaki-Okamoto transformation

Key Encapsulation Mechanism

By applying the Fujisaki-Okamoto transformation it is possible to build a IND-CCA2 KEM. Let $\mathcal{G}, \mathcal{H}, \mathcal{K}$ be hash functions and $\mathcal{E}$ an instance of HQC.PKE

- **Setup**: generate the parameters λ as in HQC.PKE, except that k will be the length of the key to be exchanged
- **KeyGen**: as in HQC.PKE

Key Encapsulation Mechanism

Let $\mathcal{G}, \mathcal{H}, \mathcal{K}$ be hash functions and $\mathcal{E}$ an instance of HQC.PKE

Encapsulation

- Generate the seed $\mathbf{m} \xleftarrow{\$} \mathbb{F}_2^k$ and the randomness $\theta \leftarrow \mathcal{G}(\mathbf{m})$
- Generate the ciphertext $\mathbf{c} \leftarrow (\mathbf{u}, \mathbf{v}) = \mathcal{E}.\text{Encrypt}(\text{pk}, \mathbf{m}, \theta)$
- Derive the symmetric key $K = \mathcal{K}(\mathbf{m}, \mathbf{c})$
- Calculate $\mathbf{d} \leftarrow \mathcal{H}(\mathbf{m})$
- Send the pair $(\mathbf{c}, \mathbf{d})$

Key Encapsulation Mechanism

Let $\mathcal{G}, \mathcal{H}, \mathcal{K}$ be hash functions and $\mathcal{E}$ an instance of HQC.PKE

Decapsulation

- Decrypt $\mathbf{m}' \leftarrow \mathcal{E}.\text{Decrypt}(\text{sk}, \mathbf{c})$
- Compute $\theta' \leftarrow \mathcal{G}(\mathbf{m}')$ and $\mathbf{c}' \leftarrow \mathcal{E}.\text{Encrypt}(\text{pk}, \mathbf{m}', \theta')$
- Check that $\mathbf{c} = \mathbf{c}'$ and $\mathbf{d} = \mathcal{H}(\mathbf{m}')$
- Derive the shared key $K \leftarrow \mathcal{K}(\mathbf{m}, \mathbf{c})$

Polynomial multiplication - implementation

Polynomial representation

We consider the ring of binary polynomials $\mathcal{R} = \mathbb{F}_2[x]/\langle x^n + 1 \rangle$

We have two ways of representing binary polynomials:

- sparse: a vector of indexes, every index is a non-null coefficient
- dense: a vector of coefficients, every entry is a coefficient

Example

The polynomial $f(x) = x^{15} + x^{13} + x^8 + x^6 + x + 1$ can be represented as:

- sparse: $[15, 13, 8, 6, 1, 0]$
- dense: $[1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1]$

Memory usage - 32 bit architectures

- sparse: one integer per coefficient (*i.e.* 4 bytes per coefficient)
- dense: one bit per coefficient (*i.e.* 0.25 bytes per coefficient)

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Shift&Add

Multiplying a dense polynomial by x^i is equivalent to a circular shift ($\lll_n$) of i positions to the left:

- $a(x)$ in dense representation
- $b(x) = [b_0, \cdots, b_{\omega-1}]$ in sparse representation
- Calculate $c_i(x) \leftarrow a(x) \lll_n b_i$
- Sum the partial results $c(x) = \sum_{i=0}^{\omega-1} c_i(x)$

Pros and cons

- Extremely fast for $\omega \ll n$
- Non constant-time execution

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Constant time S&A

Calculate all the shifts, but only sum the ones that are needed:

Monomial-Polynomial multiplication

Let $a(x)$ be a dense polynomial and x^b a monomial, with $b = (\beta_{m-1}\beta_{m-2}\cdots\beta_0)_{\texttt{bin}}$. For each bit $i \in \{0, \cdots, m-1\}$, we have:

- $tmp(x) \leftarrow a(x) \ll_n 2^i$
- if $\beta_i = 1$ then $\texttt{mask} = [111\cdots11]_n$, otherwise $\texttt{mask} = [000\cdots00]_n$
- $c(x) \leftarrow c(x) + ((tmp(x) \wedge \texttt{mask}))$
  - $\beta_i = 1$: $c(x) \leftarrow c(x) + tmp(x)$
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Frobenius Additive FFT

Based on the BIKE optimized implementation on ARM Cortex-M4 [CCK21]

Let $a(x)$ and $b(x)$ be two dense polynomials of degree n a power of 2

- Basis conversion (from monomial to novelpoly)
- Evaluation with the FFT
  - Encode (truncated FFT)
  - Butterflies
- Pointwise multiplication
- Inverse FFT
- Basis conversion (from novelpoly to monomial)

Inverse operations follow the same principles of the forward ones.

Operands have fixed sizes $\rightarrow$ unroll and optimize recursive calls

Basis conversion

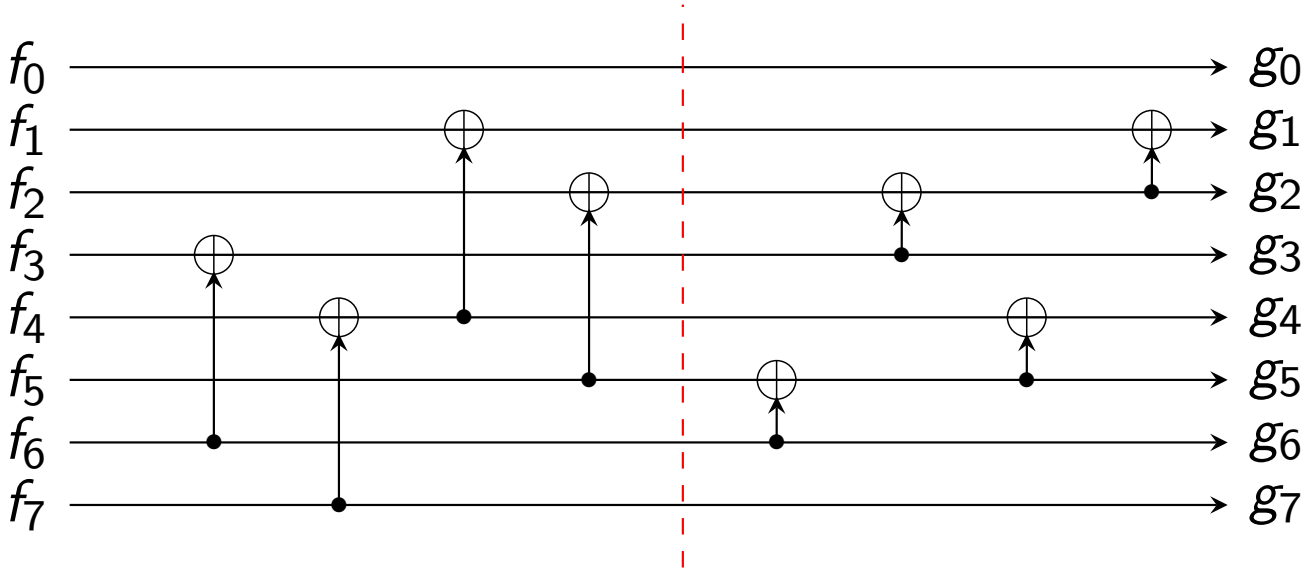
Recursively split the polynomial into halves by dividing by $s_i(x)$

No actual division required, just XORs:

Also, skip the (padded) null coefficients

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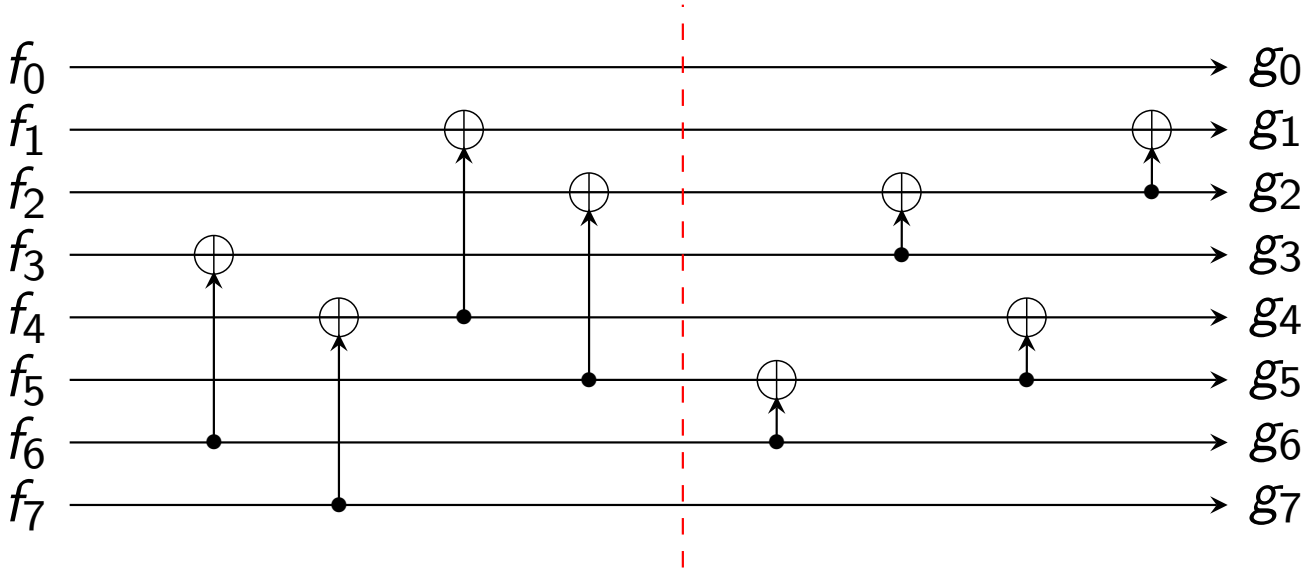
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Evaluation with the FFT

Frobenius partition

Polynomial degree $2 \cdot 17669 \rightarrow n = 65536$

32-bit words $\rightarrow m = 32$

Optimal Frobenius partition:

$$\mathbf{S} = v_{l+m/2} + \mathbf{W}_l = v_{27} + \mathbf{W}_{11}$$

with $l = \log_2(n/m)$

Encoding

The first $\log_2(m)$ layers of the FFT use multipliers depending only on m

- Encode (truncated FFT) the first $\log_2(m)$ recursion layers
- *Classic* butterflies for the remaining layers

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Encoding

The first $l_m = \log_2(m)$ layers of the FFT convert elements from $\mathbb{F}_2^n$ to $\mathbb{F}_{2^m}^{n'}$.

The twiggle factors can be rewritten as

$$s_{l+1}(v_{l+m/2}), \cdots, s_{l+l_m}(v_{l+m/2})$$

$$v_{m/2-1}, \cdots, v_{m/2-l_m}$$

so the encoding can be implemented as a linear map:

$$g_i = \sum_{j=0}^{m-1} f_{j \cdot n' + i} \cdot r_j = \begin{bmatrix} f_{0 \cdot n'} & f_{1 \cdot n'} & \cdots & f_{(m-1) \cdot n'} \end{bmatrix} \cdot \begin{bmatrix} r_0 \\ r_1 \\ \vdots \\ r_{m-1} \end{bmatrix}$$

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Multiplications

Butterfly twiggle factors

Multiplication step: $\beta \cdot p_1(s_1(\beta))$, $\beta = (v + w) \in v_{l+m/2-i} + \mathbf{W}_{l-i}$

- Precompute and store all the β values
- Notice that $\gamma \cdot \beta = \gamma \cdot (v + w) = \gamma \cdot v + \gamma \cdot w$
  - $\gamma \cdot v$ can be computed with a binary circuit (v are fixed)
  - $\gamma \cdot w$ can be computed in a subfield to reduce the number of bits

Pointwise multiplications

The resulting evaluation points are, by construction, in $\mathbb{F}_{2^{32}}$.
Apply Karatsuba (only once) to operate in $\mathbb{F}_{2^{16}}$ and have the results fit in the 32-bit CPU registers.

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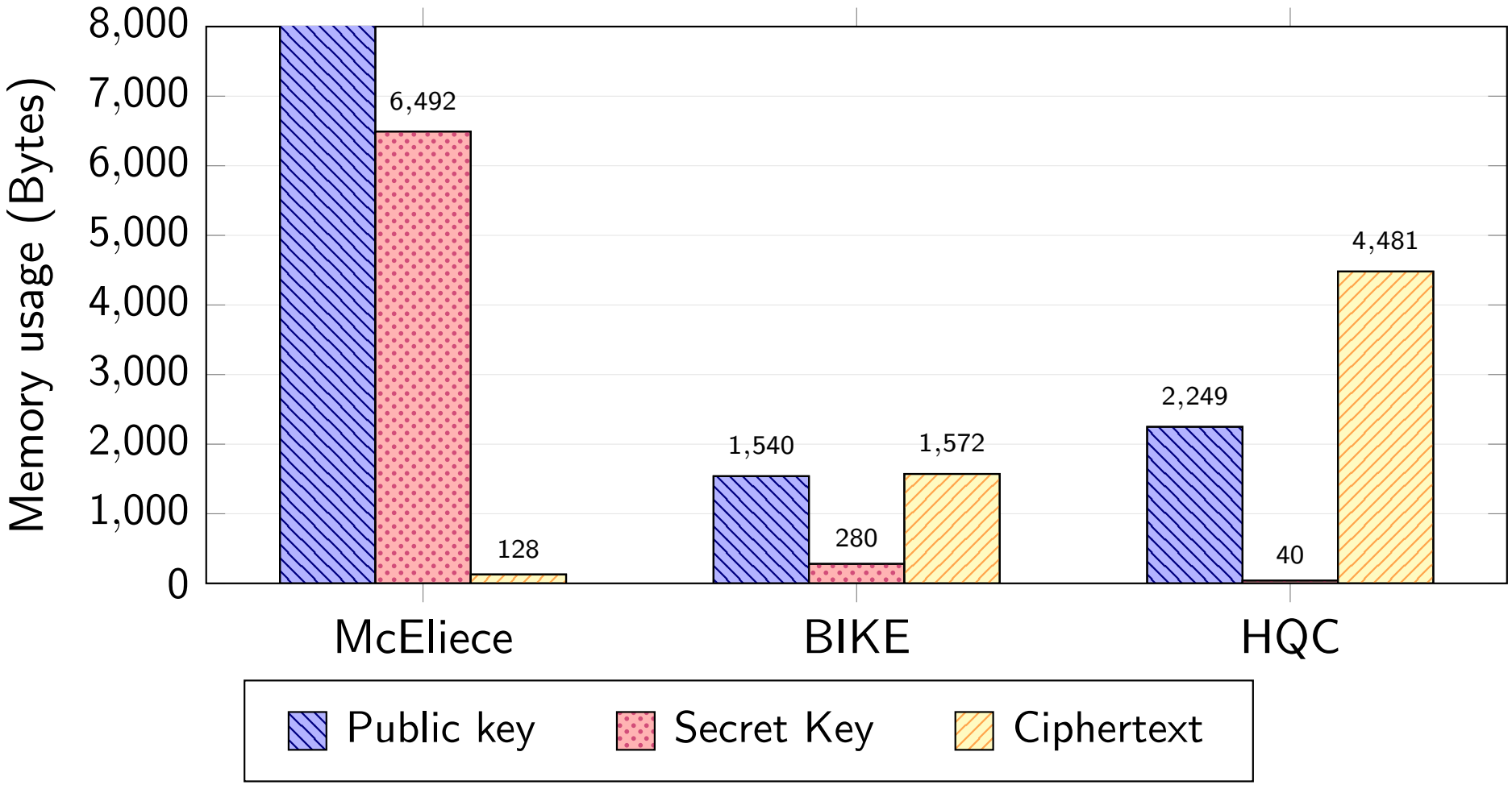
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Additional data

Constant-time: KEM

Round	Version	Mul	CT?	Encaps x86	Encaps ARM	Decaps x86	Decaps ARM
3	`reference`	Shift&Add	○	58.42	28.65	20.71	18.44
	`sparsemul`	CT Shift&Add†	•	6.75	14.29	5.81	23.70
4	`reference`	Karatsuba†	•	3.77	4.01	4.56	0.26
	`sparsemul`	CT Shift&Add†	•	1.58	2.17	1.04	0.22
	`fastmul`	Frobenius FFT	•	4.31	4.25	2.41	0.53

Memory


Memory usage (Bytes)
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
6,492
128
1,540
280
1,572
2,249
40
4,481
McEliece
BIKE
HQC
Public key
Secret Key
Ciphertext

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